Exploratory Data Analysis (EDA)

- **Exploratory Data Analysis (EDA)** is the process of examining a dataset to understand its main characteristics before performing advanced statistical

EDA is the first step in data analysis where you explore the data to answer questions such as:

- What variables are in the dataset?
- Are there missing values?
- What does the distribution of the data look like?
- Are there outliers?
- What patterns or relationships exist?

EDA helps you **understand the data before applying any statistics**.

Benefits of Exploring Data

Goals of EDA

- Identify errors or inconsistent data
- Summarize main features
- Visualize distributions and trends
- Detect outliers
- Test assumptions (normality, linearity, etc.)
- Decide on next steps: cleaning, transformation, or modelling

EDA Tools in SPSS

SPSS offers several tools to help with EDA:

a) Descriptive Statistics

- Menu: **Analyze** → **Descriptive Statistics** → **Frequencies / Descriptives**

Common outputs:

- Mean, median, mode
- Standard deviation
- Minimum, maximum
- Percentiles

Visualizations

- Menu: **Graphs** → **Chart Builder**
- Useful charts:
- Histograms (distribution)
- Boxplots (outliers)
- Bar charts
- Pie charts
- Scatterplots (relationships)

Missing Values Analysis

- Menu: **Analyze → Missing Value Analysis**

Shows:

- Patterns of missing data
- Percentage of missing values
- Possible causes or patterns

Explore Command (Most Comprehensive EDA Tool)

Menu: **Analyze** → **Descriptive Statistics** → **Explore**

Outputs include:

- Descriptive tables
- Normality tests
- Boxplots
- Stem-and-leaf plots
- Histograms
- Confidence intervals for mean

Key Steps in Conducting EDA in SPSS

Step 1: Understand Your Variables

- Identify variable types: numeric or categorical
- Check value labels and coding
- Ensure variable measurement level is correct (Nominal, Ordinal, Scale)

Step 2: Run Descriptive Statistics

Use means, medians, frequencies, and standard deviations to get a first impression of the data.

Step 3: Check Missing Data

- Look at % of missing values
- Determine if missingness is random
- Decide whether to remove or impute

Key Steps in Conducting EDA in SPSS

Step 4: Visualize Distributions

- Use histograms and boxplots to check:
- Normality
- Spread of data
- Outliers

Step 5: Identify Outliers

- Boxplots and z-scores
- Decide if outliers are genuine or errors

Step 6: Explore Relationships

- Use scatterplots or correlation (if needed) to see:
- Patterns
- Trends
- Clusters

Why EDA Is Important

- Helps avoid incorrect conclusions
- Ensures data quality
- Determines appropriate statistical tests
- Improves understanding of the dataset
- Guides cleaning, transformation, and modeling